The Journal of Open Source Software

TriPoDPy: 1D Tri-Population size distributions for Dust evolution in protoplanetary disks

Nicolas Leo Kaufmann [1], **Thomas Pfeil** [2], **Sebastian Stammler** [1], **Anna B. T. Penzlin** [1], **Sandro Christian Paetzold** [1], **and Til Birnstiel** [1,3]

1 University Observatory, Faculty of Physics, Ludwig-Maximilians-Universität München, Scheinerstr. 1, 81679 Munich, Germany **2** Center for Computational Astrophysics, Flatiron Institute, 162 Fifth Avenue, New York, NY 10010, USA **3** Exzellenzcluster ORIGINS, Boltzmannstr. 2, D-85748 Garching, Germany

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Tobias Buck

Reviewers:

- @DingshanDeng
- @wcukier
- @Sportsfan77777

Submitted: 20 November 2025
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0)

Summary

TriPoDPy is a code simulating the dust evolution, including dust growth and dynamics in protoplanetary disks using the parametric dust model presented in (Pfeil et al., 2024). The simulation evolves a dust distribution in a one-dimensional grid in the radial direction. It's written in Python and the core routines are implemented in Fortran90. The code not only solves for the evolution of the dust but also the gas disk with the canonical α-description (Shakura & Sunyaev, 1973). In addition to the original model, we added descriptions of tracers for the dust and gas, which could be used for compositional tracking of additional components.

Statement of Need

Simulating the dust evolution in protoplanetary disks, including growth and transport, is vital to understanding planet formation and the structure of protoplanetary disks. TriPoDPy describes the vertically integrated dust size distribution with a truncated power law, which allows the simulation full access to the dust size distribution, which is essential to accurately model the dust evolution and additional physical effects like photoevaporation. Additionally, TriPodPy enables the addition of tracers in gas and dust, which could be used for tracking of chemical composition, electrical charge, and other parameters. All these features make it a valuable tool for astronomers to quickly model the dust evolution of protoplanetary disks.

State of the Field

There exist multiple open-source codes that tackle the dust evolution in Protoplanetary disks by either solving the Smoluchowski Equation, e.g. Dustpy (S. M. Stammler & Birnstiel, 2022) or CuDisc (Robinson et al., 2024) or using a Monte Carlo approach (e.g. Mcdust (Vaikundaraman et al., 2025)) to simulate the mutual collisions between dust grains. However, all these simulations are computationally expensive, which calls for parametrised dust evolution models that can be used, for example, for population studies. Previous parametric models, e.g. Twopoppy (Birnstiel et al., 2012), were not designed for disks with radial sub-structures and were not calibrated for different stellar masses. These shortcomings are solved with the TriPoD Dust model.

Software design

TriPoDPy builds on the Simframe (S. Stammler & Birnstiel, 2022) framework to allow users to easily modify and use our code by handling file IO, etc. To increase performance, the core

DRAFT

routines have been implemented in `Fortran90`. Since we are solving the same problem as the full Smoluchowski solver `Dustpy`, our code inherits several shared routines, making it easy for users familiar with `Dustpy` to adopt the code and convert setups between the two codes.

Research impact statement

Currently, there are multiple ongoing projects that use this code, with the first results already in preprint (Ziampras et al., 2026). Since the code is a straight upgrade over the widely used predecessor `Twopoppy` (Birnstiel et al., 2012) (with over 700 citations) for most cases, we expect many projects to make use of our code.

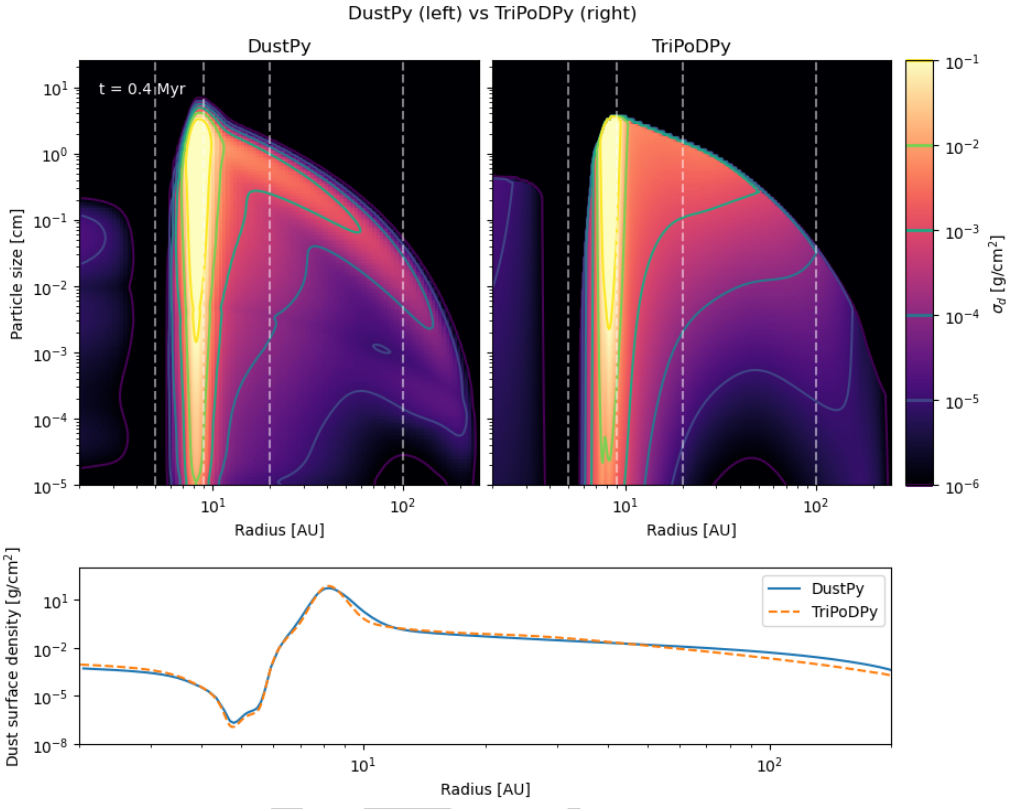
Comparison Simulation

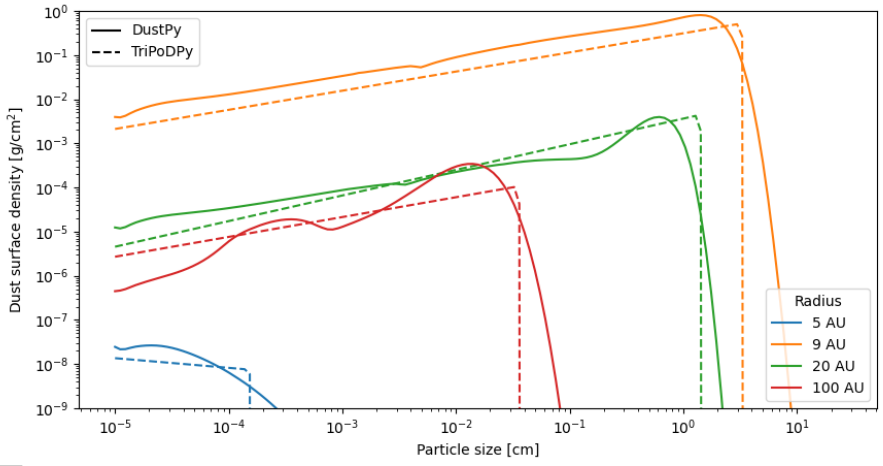
We compare a Simulation with our code with one performed with the full coagulation code `Dustpy`, illustrating how well our code performs. The parameters used for the comparison simulations can be found in the Table below:

Parameter	Value
gas surface density at 1 AU	722 $\mathrm{g/cm^2}$
temperature at 1 AU	209 K
turbulence strength (α)	10^{-3}
fragmentation velocity (v_{frag})	10 m/s
gas surface density power law p	0.85
temperature power law q	0.5
gap position	5.2 AU
$M_{\mathrm{planet}}/M_\star$	10^{-3}

We compare the particle size distribution from both simulations at 400'000 years, which can be seen in the figures below. The first figure shows the dust surface density as a function of size and radius throughout the disk (top) and the total dust surface desity as a function of radius (bottom). The second plot shows 1-D slices at different radii as indicated by the white dashed lines.

DRAFT





The `TriPoDPy` simulation runs a factor of 50 to 100 faster than the compared `DustPy` model. As we can see, the maximal sizes and dust size distributions match quite well with the full coagulation code. Since the size distribution is always assumed to be a power law, capturing multimodal distributions is not possible, as can be seen around 100 AU in the test simulation. This also affects the dust distribution on the inside of the gap, as the dust size distribution in gaps deviates from the expected power law as well. For an in-depth discussion, see (Pfeil et al., 2024).

Acknowledgments

The authors acknowledge funding from the European Union under the European Union's Horizon Europe Research and Innovation Programme 101124282 (EARLYBIRD) and funding by the

Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy - EXC-2094 - 390783311. Views and opinions expressed are, however, those of the authors only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them.

AI usage disclosure

AI (GitHub Copilot) was used to generate implementation ideas for the code, however all prompts were subsequently checked and corrected manually.

References

Birnstiel, T., Klahr, H., & Ercolano, B. (2012). A simple model for the evolution of the dust population in protoplanetary disks. *Astronomy & Astrophysics*, *539*, A148. https://doi.org/10.1051/0004-6361/201118136

Pfeil, T., Birnstiel, T., & Klahr, H. (2024). TriPoD: Tri-Population size distributions for Dust evolution: Coagulation in vertically integrated hydrodynamic simulations of protoplanetary disks. *Astronomy & Astrophysics*, *691*, A45. https://doi.org/10.1051/0004-6361/202449337

Robinson, A., Booth, R. A., & Owen, J. E. (2024). Introducing CUDISC: a 2D code for protoplanetary disc structure and evolution calculations. *Monthly Notices of the Royal Astronomical Society*, *529*(2), 1524–1541. https://doi.org/10.1093/mnras/stae624

Shakura, N. I., & Sunyaev, R. A. (1973). Black holes in binary systems. Observational appearance. *Astronomy & Astrophysics*, *24*, 337–355.

Stammler, S. M., & Birnstiel, T. (2022). DustPy: A Python Package for Dust Evolution in Protoplanetary Disks. *The Astrophysical Journal*, *935*(1), 35. https://doi.org/10.3847/1538-4357/ac7d58

Stammler, S., & Birnstiel, T. (2022). Simframe: A Python Framework for Scientific Simulations. *The Journal of Open Source Software*, *7*(69), 3882. https://doi.org/10.21105/joss.03882

Vaikundaraman, V., Gurrutxaga, N., & Drążkowska, J. (2025). mcdust: A 2D Monte Carlo code for dust coagulation in protoplanetary disks. *arXiv e-Prints*, arXiv:2507.21239. https://doi.org/10.48550/arXiv.2507.21239

Ziampras, A., Birnstiel, T., Kaufmann, N., Cecil, M., & Pfeil, T. (2026). *Planet formation at the inner edge of the dead zone – i: The interplay between accretion outbursts and dust growth*. https://arxiv.org/abs/2602.20283

DRAFT